

VidVaani

Automated, low-cost dubbing of English technical lectures into Hindi

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The problem: technical education is locked in English

- India's best engineering lectures — NPTEL alone hosts **50,000+ hours** — are overwhelmingly in English.
- A large fraction of our students think and learn in Hindi and other Indian languages; a fast English lecture is a barrier, not a resource.
- **NEP 2020** explicitly calls for higher education content in Indian languages.
- Manual dubbing is slow and expensive: NPTEL's human-verified translation effort has taken years to cover a subset of courses in 11 languages.

Goal: given any lecture video, produce a natural-sounding Hindi version in minutes, for rupees — with the technical vocabulary students actually use.

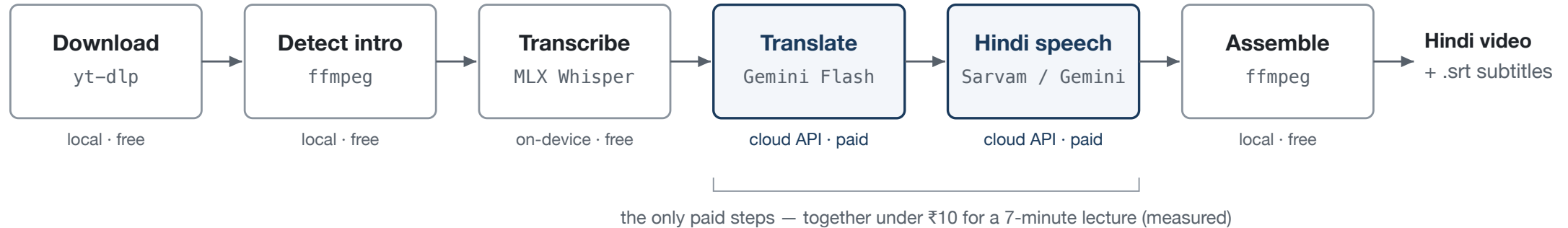
Existing options do not fit the lecture use case

Option	Cost	Why it falls short
YouTube auto-dubbing	Free	Only the channel owner can enable it; dubs are uneditable; no control over voice or terminology; widely criticised as robotic
Commercial dubbing SaaS (ElevenLabs, Rask, HeyGen, Dubverse, Murf)	₹3,000–11,000 per lecture	Priced for short marketing clips; monthly minute caps; each extra language billed again
Azure Video Translation	≈ ₹2,000 per lecture	Cheapest managed option, but limited voice choice and no terminology control
NPTEL / Bhashini human pipelines	Government-funded	High quality but slow, course-by-course; not self-serve

The gap: a self-serve tool an instructor can point at *any* video, with control over voice and vocabulary, at commodity API prices.

VidVaani: a six-stage open pipeline

Input: YouTube URL



- Single command: `vidvaani dub <URL> --full -b sarvam -v abhilash`
- Transcription runs **on-device** (MLX Whisper on Apple Silicon) — audio never leaves the machine until translation.
- Every intermediate artifact is cached: re-dubbing with a new voice skips download, transcription, and translation.

Stage design choices

Stage	Tool	Design choice
Download	<code>yt-dlp</code>	Works on any YouTube URL or local file — no allowlist
Intro detection	<code>ffmpeg silencedetect</code>	Music/intro is preserved , dubbing starts at first speech
Transcription	MLX Whisper (<code>distil-large-v3</code>)	Local, free, time-stamped segments (grouped to ≤ 15 s)
Translation	Gemini 2.5 Flash	Segment-batched, JSON-structured, duration-aware prompt
Speech synthesis	Sarvam <code>bulbul:v2</code> / Gemini TTS / Edge TTS	Native Indian voices; parallel synthesis with per-segment caching
Assembly	<code>ffmpeg</code>	Duration matching + original audio preserved during pauses

Translation that respects the clock — and the classroom

Constraints in the translation prompt: *speakable in the same duration; conversational (not literary) Hindi; technical terms stay in English.*

	Original (English)	VidVaani (Hindi)
0:00	"Hello everyone, welcome to lecture 1 of CS7015 which is the course on deep learning ."	"सभी को नमस्कार, CS7015 के पहले लेक्चर में आपका स्वागत है, जो डीप लर्निंग का कोर्स है।"
0:30	"...we hear the terms artificial neural networks , artificial neurons quite often these days."	"...आजकल हम ' आर्टिफिशियल न्यूरल नेटवर्क ', ' आर्टिफिशियल न्यूरॉन्स ' जैसे शब्द अक्सर सुनते हैं।"

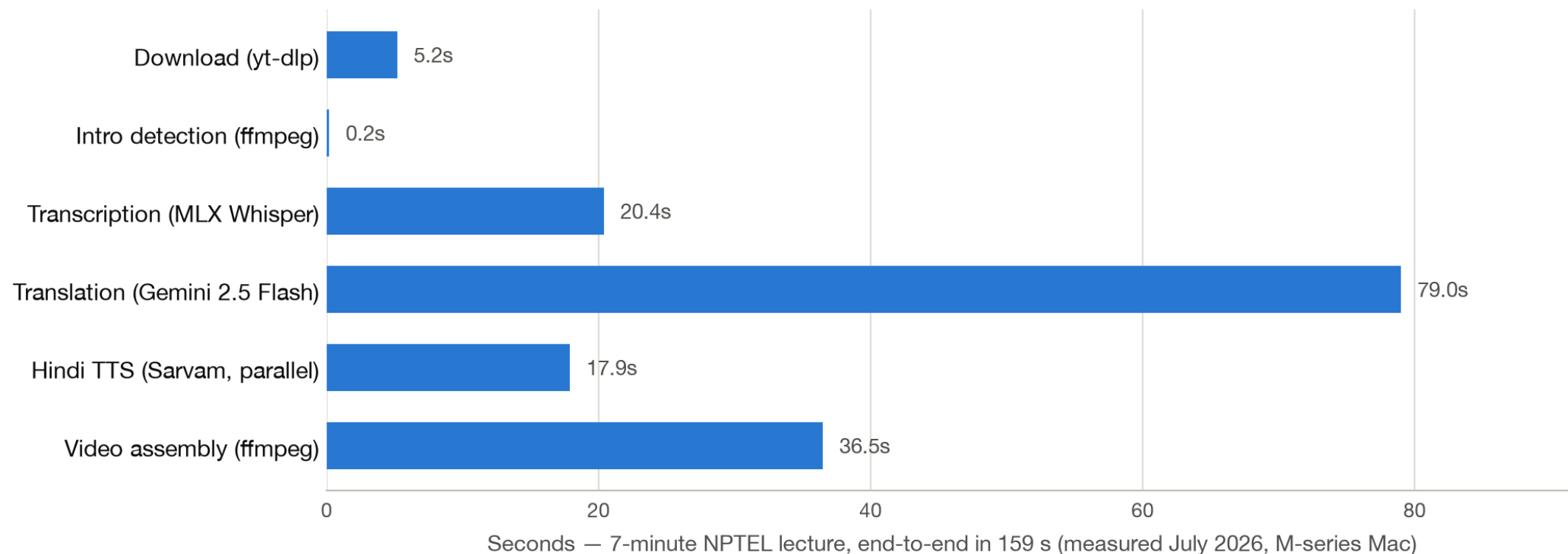
Students say *gradient* and *neural network* — a dub that renders them as प्रवणता and तंत्रिका जाल (as fully-literal systems do) sounds foreign to the very audience it serves.

Fitting Hindi speech into English timing

Hindi runs ~15–35% longer than English. Per segment, following the automatic-dubbing literature:

1. The translator gets a **word budget** ($\approx 1.8 \times$ seconds, ~ 110 wpm spoken Hindi) — fixing length at translation time beats stretching audio afterwards.
2. Synthesize at natural pace, trim trailing silence, measure with `ffprobe`.
3. If it fits ($\pm 5\%$) — done. Otherwise speed-adjust with an **asymmetric clamp (0.95x–1.35x)**: listeners tolerate faster speech far better than slowed speech.
4. Still too long? The clip **spills into the trailing pause** rather than being cut — professional dubs sacrifice timing, never content.
5. During pauses, the **original soundtrack plays** — atmosphere and intro music survive the dub.

Measured performance — faster than real time



A 7-minute lecture dubs in 2½–4 minutes end-to-end on a laptop — the bottleneck is the translation API, not synthesis. Re-runs with a different voice take under a minute (cached transcript + translation).

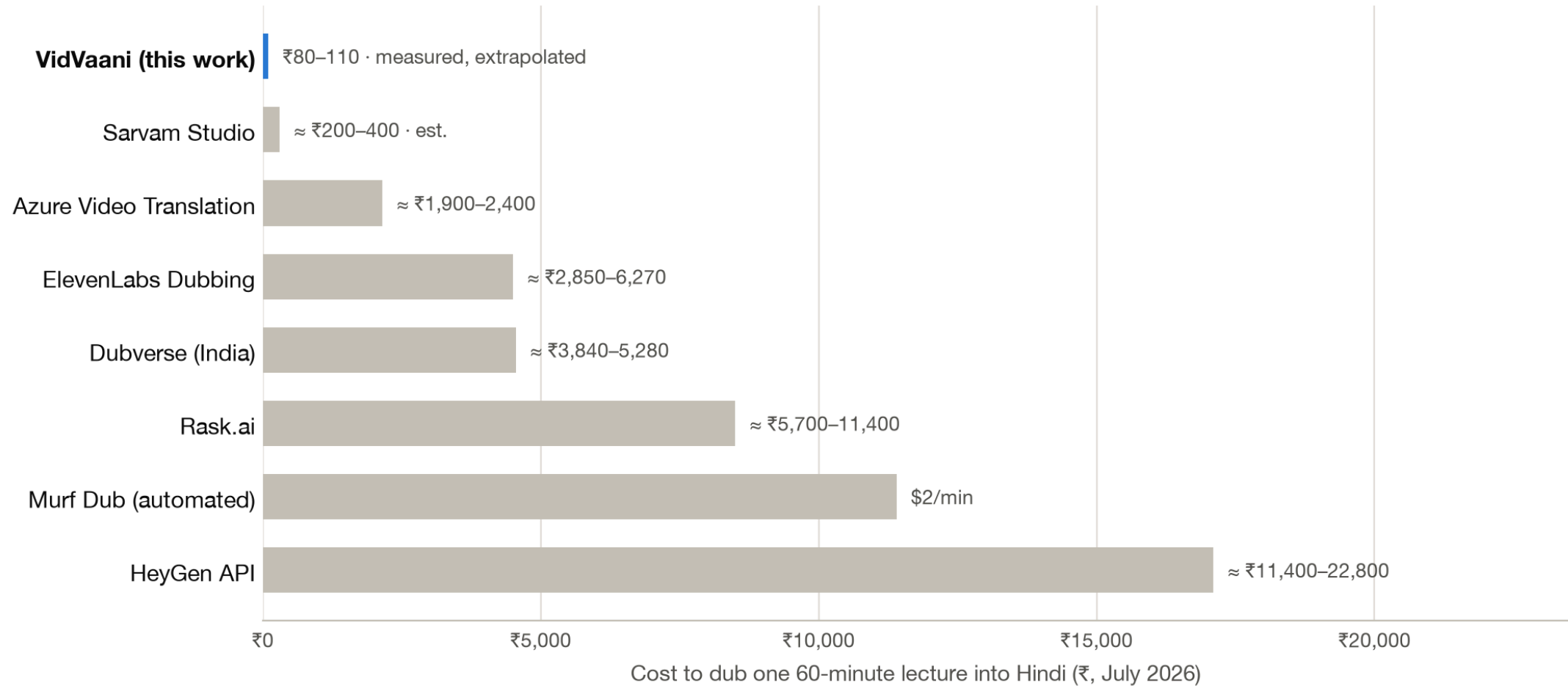
Measured cost — rupees, not lakhs

Measured on the same 7-minute lecture (5,307 Hindi characters):

	Translation	Hindi TTS	Total (7 min)	Extrapolated, 1 hour
Sarvam <code>bulbul:v2</code>	₹1.4	₹8.0	₹9.4 (\$0.10)	≈ ₹80
Gemini 2.5 Flash TTS	₹1.4	₹9.6	₹11.0 (\$0.12)	≈ ₹95
Edge TTS (free tier)	₹1.4	₹0	₹1.4	≈ ₹12

- A full **40-lecture course** ≈ **₹3,200–4,400** — less than one SaaS-dubbed lecture.
- Costs measured by the pipeline itself from API-reported token counts, at official July 2026 prices.
- Conservative: after adding word budgets to the translation, the same lecture re-ran at **₹6.7**.

Where this sits in the market



Midpoints of published prices, single target language, July 2026. YouTube auto-dubbing is free but channel-owner-only with no editorial control. Papercup/RWS (human-in-loop enterprise) quotes custom prices well above this scale.

Why not just use the platforms?

YouTube auto-dubbing

- Now rolled out to all channels — **but only the uploader** can enable it.
- A university cannot dub NPTEL or MIT content it does not own.
- Dubs cannot be edited — a mistranslated technical term can only be unpublished, not fixed.

Sarvam Studio & SaaS dubbers

- Produce formal, fully-translated Hindi (क्षेत्रफल for "area") — jarring for STEM learners used to Hinglish.
- No control over segment-level phrasing or terminology.
- VidVaani uses the **same Sarvam voices** with full editorial control, at API prices.

Notably, the popular open-source dubbing tools (pyvideotrans ★18k, VideoLingo ★17k) integrate **no Indian-language TTS at all** — VidVaani fills a real gap.

What a run looks like

```
$ vidvaani dub "https://youtube.com/watch?v=4TC5s_xNKSs" --full -b sarvam -v anushka
```

Downloaded: Deep Learning(CS7015): Lec 1.1 Biological Neuron

Transcribed: 33 segments

Generated subtitles: 4TC5s_xNKSs_hindi.srt

TTS: 33/33 segments (parallel)

Created: 4TC5s_xNKSs_hindi_anushka.mp4

Timing Breakdown

Download	5.2s	3.3%
Transcription	20.4s	12.8%
Translation	79.0s	49.6%
TTS Generation	17.9s	11.2%
Video Assembly	36.5s	22.9%
Total	159.2s	100%

API Costs

Translation	9,542 tokens	\$0.015
TTS (Sarvam)	5,307 chars	Rs 7.96
Total		Rs 9.4

Every run reports its own timing and cost — the numbers on the previous slides are this output, not estimates.

Demonstrations

nipunbatra.github.io/vidvaani — plays in any browser:

1. **58-second calculus clip, five ways** — original, Sarvam and Gemini voices, plus Sarvam's own Dashboard output for direct comparison.
2. **Full 7-minute NPTEL lecture** — original beside Hindi dub; intro music preserved; five voices; burned-in subtitles.

What to listen for:

- Technical terms stay in English (*deep learning*, CS7015).
- Pauses land where the professor pauses.
- Native Indian prosody (Sarvam) vs. slight Western accent (Gemini).

Acknowledgements

- Most of this material is based on the article “Deep Learning in Neural Networks: An Overview” by J. Schmidhuber^[1]
- The errors, if any, are due to me and I apologize for them
- Feel free to contact me if you think certain portions need to be corrected (please provide appropriate references)

कुछ आधार और डिसक्लेमर। मटेरियल पहले बुलेटेड पेपर से लिया है। यह अट्ठारह सौ इकहत्तर से है, तो कुछ गलतियाँ हो सकती हैं। शायद कुछ तथ्य गलत हों।

Dubbed lecture with generated Hindi subtitles burned in.



Scan to open the demo page.

Honest limitations

- **No human review loop yet** — NPTEL's own effort treats faculty verification as essential for pedagogy; ₹80/lecture buys the draft, not the sign-off.
- **Timing fit is still partly post-hoc** — word budgets plus a 0.95–1.35× speed clamp handle most segments, but very dense passages can still sound hurried (see next slide).
- Gemini TTS models are **preview** — pricing and behaviour may change; Edge TTS is an unofficial free endpoint, fine for prototyping only.
- No lip-sync and no voice cloning of the original lecturer (the latter also needs consent workflows).
- Hindi first; Sarvam voices cover 11 Indian languages, so extension is natural.

Roadmap: smarter time alignment

We surveyed the automatic-dubbing literature (Amazon, Microsoft, IIT Madras, IWSLT) and re-based the alignment design on it — full notes in `docs/timing-alignment.md`:

Already implemented — word budgets in the translation prompt; asymmetric speed clamp; spill-into-pause instead of truncation; trailing-silence trimming.

Next:

1. **TTS-native pace instead of waveform stretching** — Sarvam exposes `pace` (0.3–3.0×); synthesis at the right speed beats resampling in every published comparison.
2. **Re-translate outliers once** — segments still needing $> 1.3\times$ speed get one "shorten by 25%" LLM pass; ~25% of En→Hi segments are expected to qualify.
3. **Sentence-boundary segmentation** with word-level timestamps ($\approx +4.5$ BLEU over blind packing).
4. **Terminology glossary** carried across translation batches for consistency.
5. **Evaluation study** — student MOS ratings and comprehension quizzes, Hindi dub vs. English original.

Summary

- **Any lecture** → **natural Hindi dub in minutes**: one command, six automated stages, technical vocabulary preserved.
- **Measured, not estimated**: a 7-min lecture in ~3 minutes for under ₹10; $\approx$ ₹80–110 per lecture-hour; a 40-lecture course for the price of a textbook.
- **20–100× cheaper** than commercial dubbing platforms, with editorial control none of them offer.
- Open source (MIT): github.com/nipunbatra/vidvaani
- Live demos: nipunbatra.github.io/vidvaani

Next: faculty-in-the-loop review, more Indian languages, and a student comprehension study.